SPETRO GEOPORTAL ACADEMY | OFFICIAL COURSE HANDBOOK

QGIS 3.34 User Guide & Training Manual

Author: QGIS Association & Documentation Team

Scope: Practical Field Manual & Reference Guide | Pages: 12

Course Overview & Objectives

Comprehensive practical guide covering GIS interface, vector and raster layers, advanced cartography, spatial analysis algorithms, and Python console automation.

MODULE OUTLINE:

- **1. Introduction to QGIS**
- **2. Vector Data & Geoprocessing**
- **3. Raster Analysis & Terrain Modeling**
- **4. Professional Cartography & Map Layouts**
- **5. Python Automation (PyQGIS)**

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1. Introduction to QGIS

Overview of Quantum GIS architecture, coordinate reference systems (CRS - WGS84, EPSG:4326, UTM 35S / 36S for East Africa), interface navigation, layer panel, and map canvas rendering.

KEY TAKEAWAYS & PRACTICAL LAB INSTRUCTIONS

1. Load corresponding dataset layers from the SPETRO Data Harvester or Hugging Face.
2. Execute geoprocessing algorithms and verify outputs in the Interactive Map Editor.
3. Export cartographic layouts, GeoTIFF rasters, or GeoJSON vector polygons for field use.

2. Vector Data & Geoprocessing

Vector geometries: points, linestrings, polygons. Attribute table management, SQL filtering, field calculator, geometry generation, spatial joins, buffer analysis, clipping, and dissolve.

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3. Raster Analysis & Terrain Modeling

Digital Elevation Models (DEM), hillshade generation, slope calculation, aspect derivation, contour line extraction, raster calculator expressions, and multi-band satellite raster styling.

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4. Professional Cartography & Map Layouts

Print layout composer, map scales, north arrows, dynamic legends, multi-page spatial atlas generation, and high-resolution PDF/GeoTIFF export standards.

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3. Export cartographic layouts, GeoTIFF rasters, or GeoJSON vector polygons for field use.

5. Python Automation (PyQGIS)

Using the Python console in QGIS, writing batch processing scripts, automated layer styling, and headless processing algorithms using QgsProcessing.

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